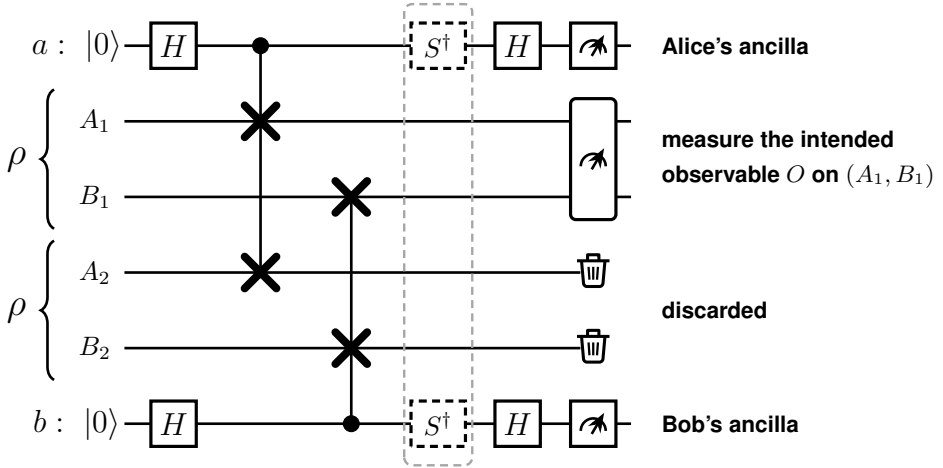


run 1 ($S^\dagger$ omitted): $C_{XX}(O) = \langle X_a X_b \otimes O_{A_1 B_1} \rangle$

run 2 ($S^\dagger$ applied): $C_{YY}(O) = \langle Y_a Y_b \otimes O_{A_1 B_1} \rangle$

$$\Delta(O) = C_{XX}(O) - C_{YY}(O) = \mathrm{Tr}(O\rho^2), \qquad \langle O \rangle_{\mathrm{VD}} = \frac{\Delta(O)}{\Delta(I)} = \frac{\mathrm{Tr}(O\rho^2)}{\mathrm{Tr}\,\rho^2}$$

Both runs are needed — a single XX setting is not the estimator. Alice holds $\{a, A_1, A_2\}$ and Bob holds $\{b, B_1, B_2\}$: no two-qubit gate crosses the cut.